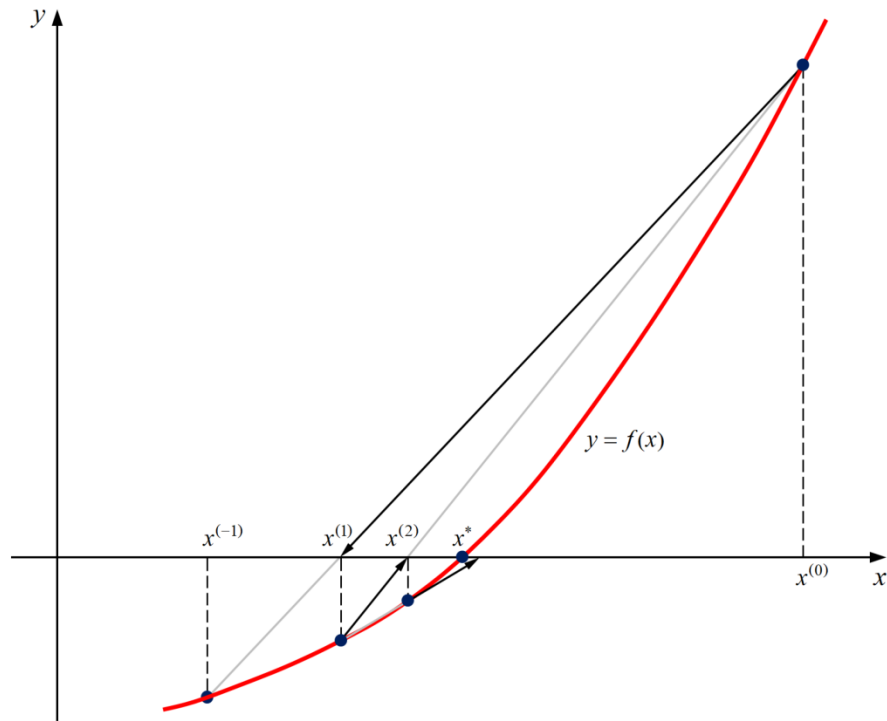


Secant Method



Newton-Raphson Method:

$$x^{(k)} = x^{(k-1)} - \frac{f(x^{(k-1)})}{f'(x^{(k-1)})}$$

Derivative Approximation:

$$f'(x^{(k-1)}) \cong \frac{f(x^{(k-1)}) - f(x^{(k-2)})}{x^{(k-1)} - x^{(k-2)}}$$

Recurrence Relation:

$$x^{(k)} = x^{(k-1)} - \frac{x^{(k-1)} - x^{(k-2)}}{f(x^{(k-1)}) - f(x^{(k-2)})} \cdot f(x^{(k-1)})$$

Relative Error:

$$\varepsilon^{(k)} = \frac{|x^{(k)} - x^{(k-1)}|}{|x^{(k)}|}$$

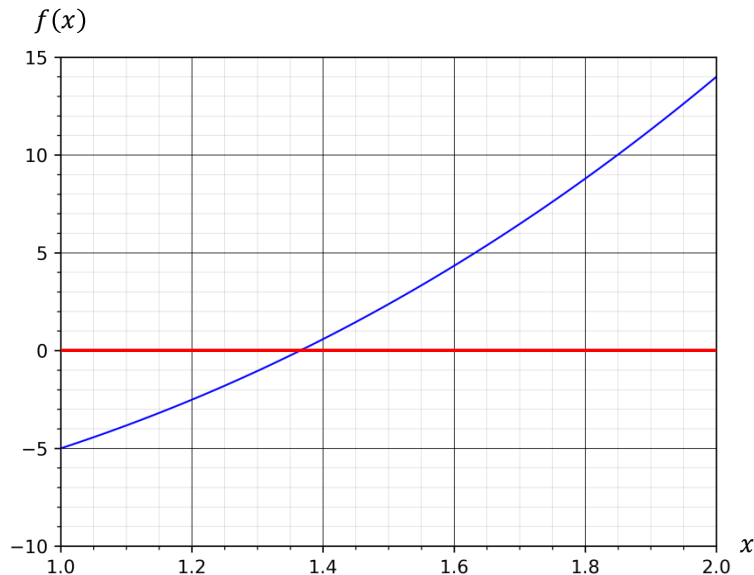
Relative Residual:

$$\varepsilon^{(k)} = \frac{|r^{(k)}|}{|r^{(0)}|} = \frac{|f(x^{(k)})|}{|f(x^{(0)})|}$$

Secant Method

Example 02-05: Find a root of equation $x^3 + 4x^2 - 10 = 0$ using secant method with initial guess $x^{(-1)} = 1, x^{(0)} = 2$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$x^{(k)} = x^{(k-1)} - \frac{x^{(k-1)} - x^{(k-2)}}{f(x^{(k-1)}) - f(x^{(k-2)})} \cdot f(x^{(k-1)}); \quad f(x) = x^3 + 4x^2 - 10$$



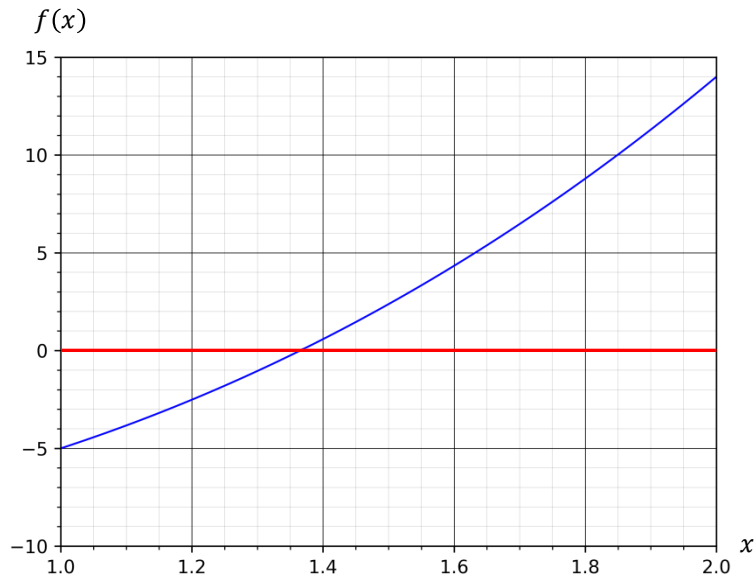
Secant Method

Example 02-05: Find a root of equation $x^3 + 4x^2 - 10 = 0$ using secant method with initial guess $x^{(-1)} = 1, x^{(0)} = 2$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$x^{(k)} = x^{(k-1)} - \frac{x^{(k-1)} - x^{(k-2)}}{f(x^{(k-1)}) - f(x^{(k-2)})} \cdot f(x^{(k-1)}); \quad f(x) = x^3 + 4x^2 - 10$$

$k = 1$

$$x^{(-1)} = 1, x^{(0)} = 2$$



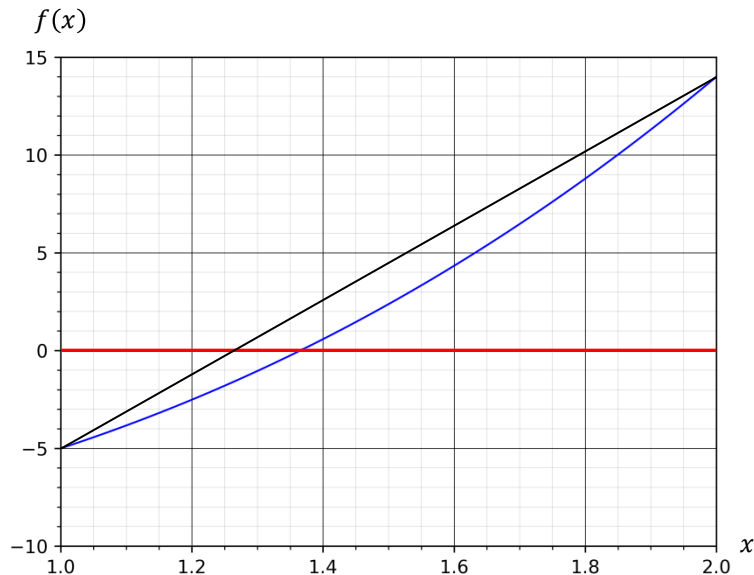
Secant Method

Example 02-05: Find a root of equation $x^3 + 4x^2 - 10 = 0$ using secant method with initial guess $x^{(-1)} = 1, x^{(0)} = 2$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$x^{(k)} = x^{(k-1)} - \frac{x^{(k-1)} - x^{(k-2)}}{f(x^{(k-1)}) - f(x^{(k-2)})} \cdot f(x^{(k-1)}); \quad f(x) = x^3 + 4x^2 - 10$$

$k = 1$

$$x^{(-1)} = 1, x^{(0)} = 2$$



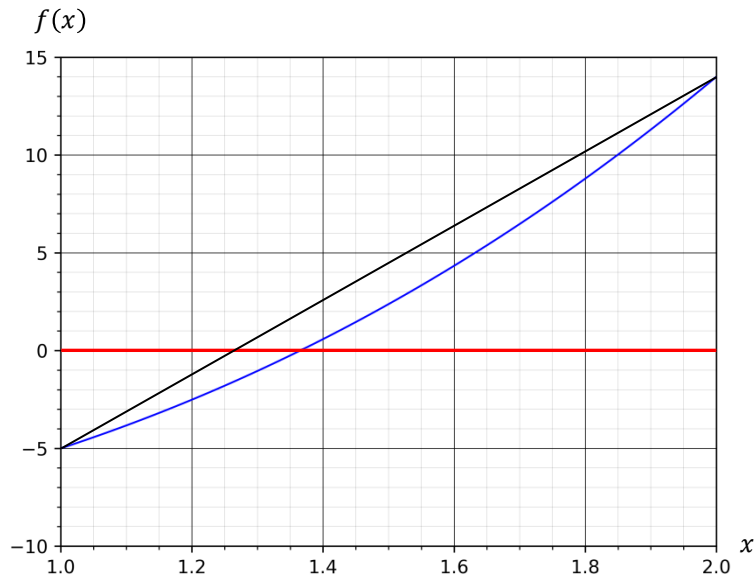
Secant Method

Example 02-05: Find a root of equation $x^3 + 4x^2 - 10 = 0$ using secant method with initial guess $x^{(-1)} = 1, x^{(0)} = 2$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$x^{(k)} = x^{(k-1)} - \frac{x^{(k-1)} - x^{(k-2)}}{f(x^{(k-1)}) - f(x^{(k-2)})} \cdot f(x^{(k-1)}); \quad f(x) = x^3 + 4x^2 - 10$$

$k = 1$

$$x^{(-1)} = 1, x^{(0)} = 2 \rightarrow x^{(1)} = 2 - \frac{2 - 1}{f(2) - f(1)} \cdot f(2)$$



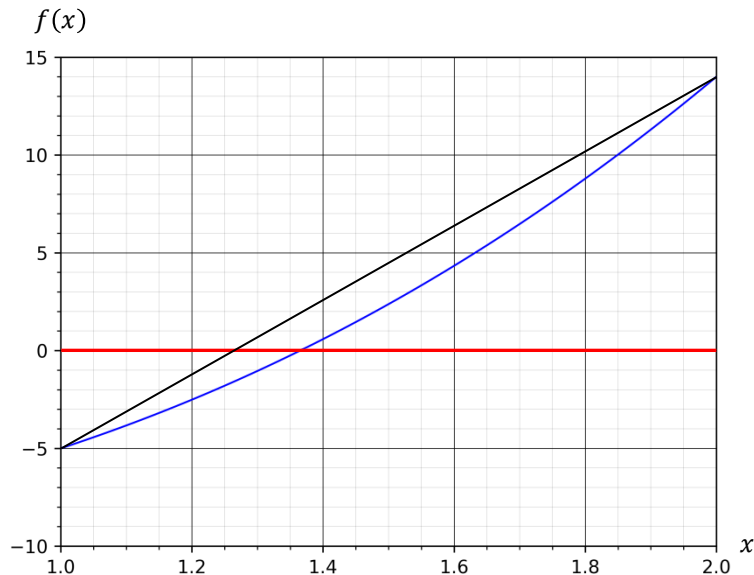
Secant Method

Example 02-05: Find a root of equation $x^3 + 4x^2 - 10 = 0$ using secant method with initial guess $x^{(-1)} = 1, x^{(0)} = 2$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$x^{(k)} = x^{(k-1)} - \frac{x^{(k-1)} - x^{(k-2)}}{f(x^{(k-1)}) - f(x^{(k-2)})} \cdot f(x^{(k-1)}); \quad f(x) = x^3 + 4x^2 - 10$$

$k = 1$

$$x^{(-1)} = 1, x^{(0)} = 2 \rightarrow x^{(1)} = 2 - \frac{2 - 1}{14 - (-5)} \cdot (14)$$



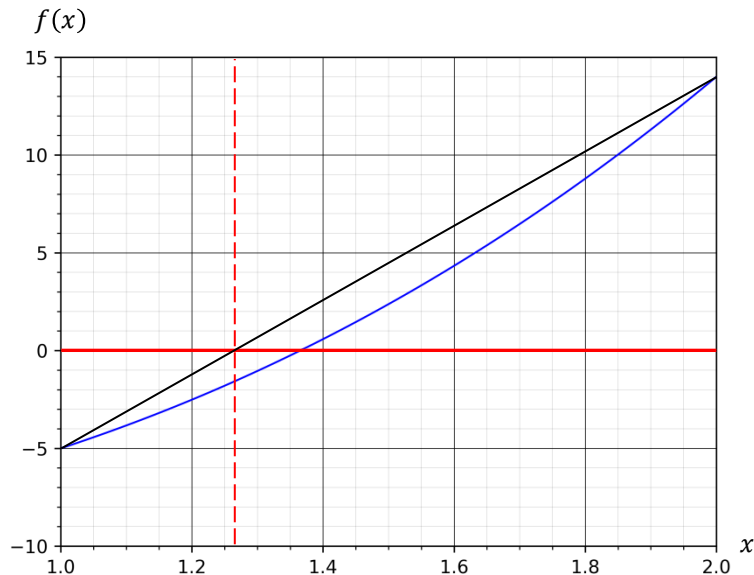
Secant Method

Example 02-05: Find a root of equation $x^3 + 4x^2 - 10 = 0$ using secant method with initial guess $x^{(-1)} = 1, x^{(0)} = 2$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$x^{(k)} = x^{(k-1)} - \frac{x^{(k-1)} - x^{(k-2)}}{f(x^{(k-1)}) - f(x^{(k-2)})} \cdot f(x^{(k-1)}); \quad f(x) = x^3 + 4x^2 - 10$$

$k = 1$

$$x^{(-1)} = 1, x^{(0)} = 2 \rightarrow x^{(1)} = 2 - \frac{2 - 1}{14 - (-5)} \cdot (14) = 1.263158$$



Secant Method

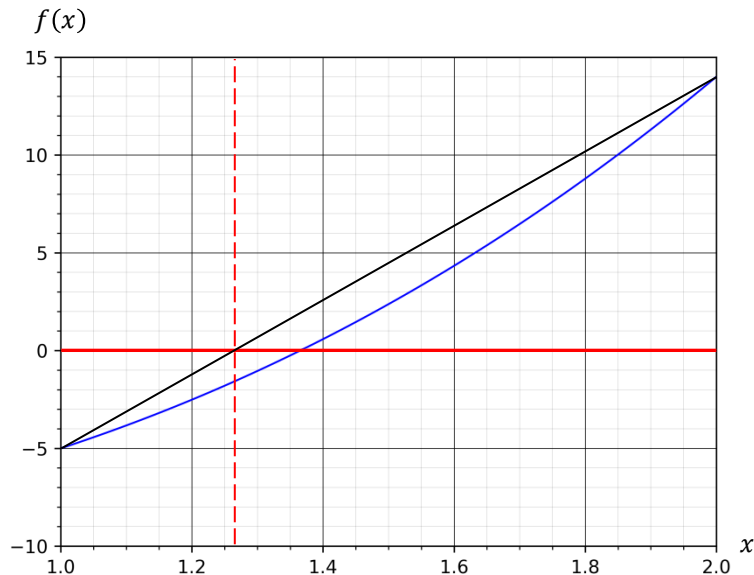
Example 02-05: Find a root of equation $x^3 + 4x^2 - 10 = 0$ using secant method with initial guess $x^{(-1)} = 1, x^{(0)} = 2$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$x^{(k)} = x^{(k-1)} - \frac{x^{(k-1)} - x^{(k-2)}}{f(x^{(k-1)}) - f(x^{(k-2)})} \cdot f(x^{(k-1)}); \quad f(x) = x^3 + 4x^2 - 10$$

$k = 1$

$$x^{(-1)} = 1, x^{(0)} = 2 \rightarrow x^{(1)} = 2 - \frac{2 - 1}{14 - (-5)} \cdot (14) = 1.263158$$

$$\varepsilon^{(1)} = \frac{|x^{(1)} - x^{(0)}|}{|x^{(1)}|} = \frac{|1.263158 - 2|}{|1.263158|} = 0.583333 > \varepsilon_t$$



Secant Method

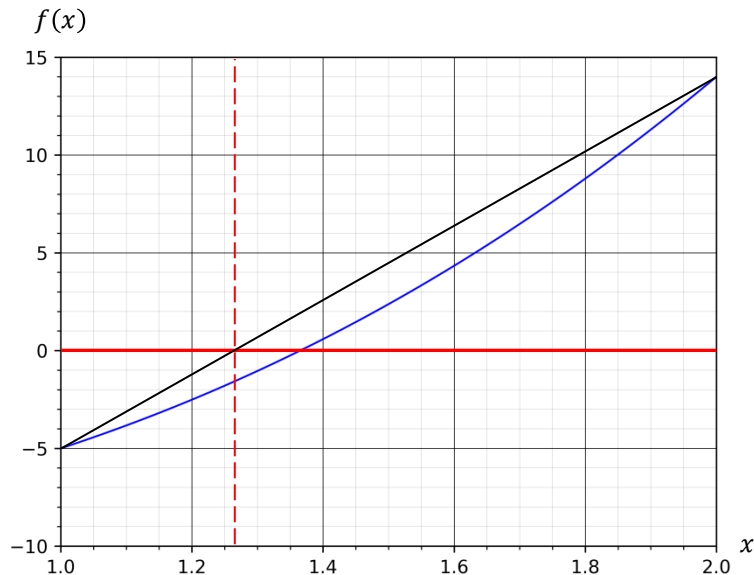
Example 02-05: Find a root of equation $x^3 + 4x^2 - 10 = 0$ using secant method with initial guess $x^{(-1)} = 1, x^{(0)} = 2$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$x^{(k)} = x^{(k-1)} - \frac{x^{(k-1)} - x^{(k-2)}}{f(x^{(k-1)}) - f(x^{(k-2)})} \cdot f(x^{(k-1)}); \quad f(x) = x^3 + 4x^2 - 10$$

$k = 1$

$$x^{(-1)} = 1, x^{(0)} = 2 \rightarrow x^{(1)} = 2 - \frac{2 - 1}{14 - (-5)} \cdot (14) = 1.263158$$

$$\varepsilon^{(1)} = \frac{|x^{(1)} - x^{(0)}|}{|x^{(1)}|} = \frac{|1.263158 - 2|}{|1.263158|} = 0.583333 > \varepsilon_t \rightarrow \text{continue}$$



Secant Method

Example 02-05: Find a root of equation $x^3 + 4x^2 - 10 = 0$ using secant method with initial guess $x^{(-1)} = 1, x^{(0)} = 2$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$x^{(k)} = x^{(k-1)} - \frac{x^{(k-1)} - x^{(k-2)}}{f(x^{(k-1)}) - f(x^{(k-2)})} \cdot f(x^{(k-1)}); \quad f(x) = x^3 + 4x^2 - 10$$

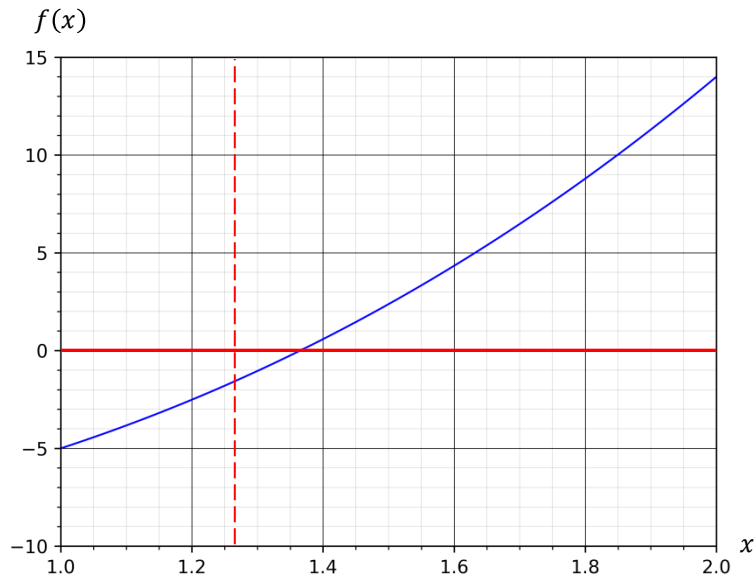
$k = 1$

$$x^{(-1)} = 1, x^{(0)} = 2 \rightarrow x^{(1)} = 2 - \frac{2 - 1}{14 - (-5)} \cdot (14) = 1.263158$$

$$\varepsilon^{(1)} = \frac{|x^{(1)} - x^{(0)}|}{|x^{(1)}|} = \frac{|1.263158 - 2|}{|1.263158|} = 0.583333 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$x^{(0)} = 2, x^{(1)} = 1.263158$$



Secant Method

Example 02-05: Find a root of equation $x^3 + 4x^2 - 10 = 0$ using secant method with initial guess $x^{(-1)} = 1, x^{(0)} = 2$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$x^{(k)} = x^{(k-1)} - \frac{x^{(k-1)} - x^{(k-2)}}{f(x^{(k-1)}) - f(x^{(k-2)})} \cdot f(x^{(k-1)}); \quad f(x) = x^3 + 4x^2 - 10$$

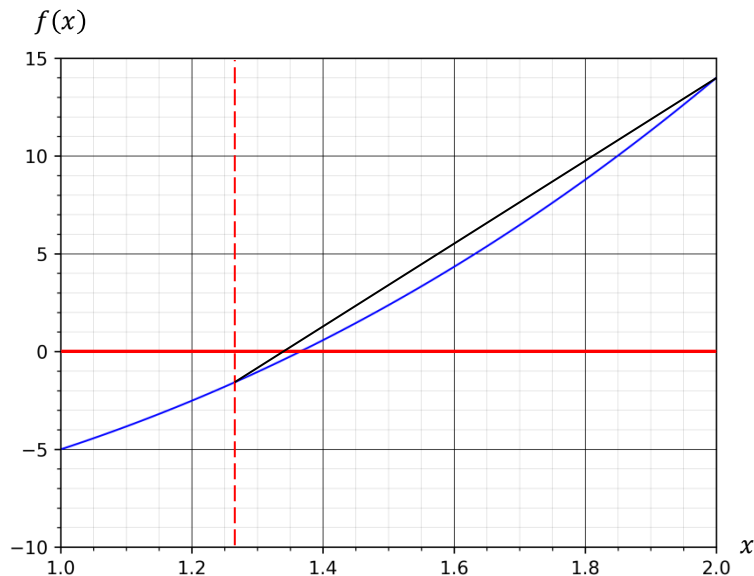
$k = 1$

$$x^{(-1)} = 1, x^{(0)} = 2 \rightarrow x^{(1)} = 2 - \frac{2 - 1}{14 - (-5)} \cdot (14) = 1.263158$$

$$\varepsilon^{(1)} = \frac{|x^{(1)} - x^{(0)}|}{|x^{(1)}|} = \frac{|1.263158 - 2|}{|1.263158|} = 0.583333 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$x^{(0)} = 2, x^{(1)} = 1.263158$$



Secant Method

Example 02-05: Find a root of equation $x^3 + 4x^2 - 10 = 0$ using secant method with initial guess $x^{(-1)} = 1, x^{(0)} = 2$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$x^{(k)} = x^{(k-1)} - \frac{x^{(k-1)} - x^{(k-2)}}{f(x^{(k-1)}) - f(x^{(k-2)})} \cdot f(x^{(k-1)}); \quad f(x) = x^3 + 4x^2 - 10$$

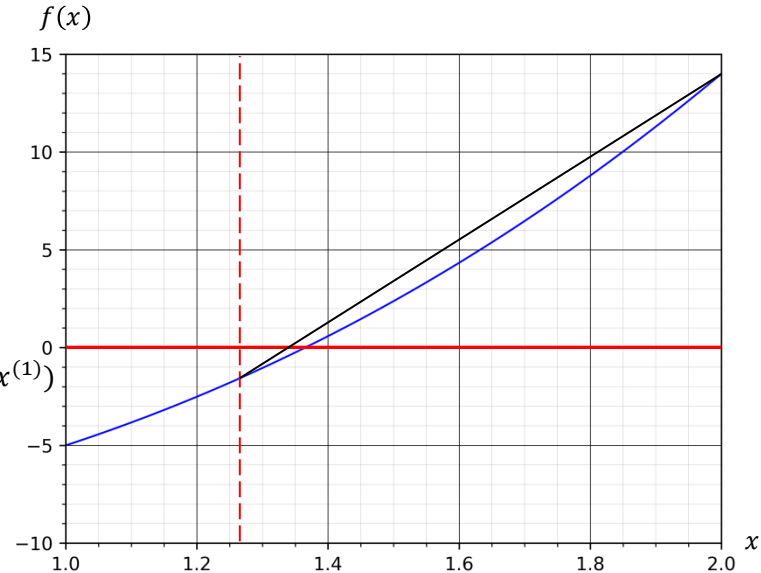
$k = 1$

$$x^{(-1)} = 1, x^{(0)} = 2 \rightarrow x^{(1)} = 2 - \frac{2 - 1}{14 - (-5)} \cdot (14) = 1.263158$$

$$\varepsilon^{(1)} = \frac{|x^{(1)} - x^{(0)}|}{|x^{(1)}|} = \frac{|1.263158 - 2|}{|1.263158|} = 0.583333 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$x^{(0)} = 2, x^{(1)} = 1.263158 \rightarrow x^{(2)} = x^{(1)} - \frac{x^{(1)} - x^{(0)}}{f(x^{(1)}) - f(x^{(0)})} \cdot f(x^{(1)})$$



Secant Method

Example 02-05: Find a root of equation $x^3 + 4x^2 - 10 = 0$ using secant method with initial guess $x^{(-1)} = 1, x^{(0)} = 2$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$x^{(k)} = x^{(k-1)} - \frac{x^{(k-1)} - x^{(k-2)}}{f(x^{(k-1)}) - f(x^{(k-2)})} \cdot f(x^{(k-1)}); \quad f(x) = x^3 + 4x^2 - 10$$

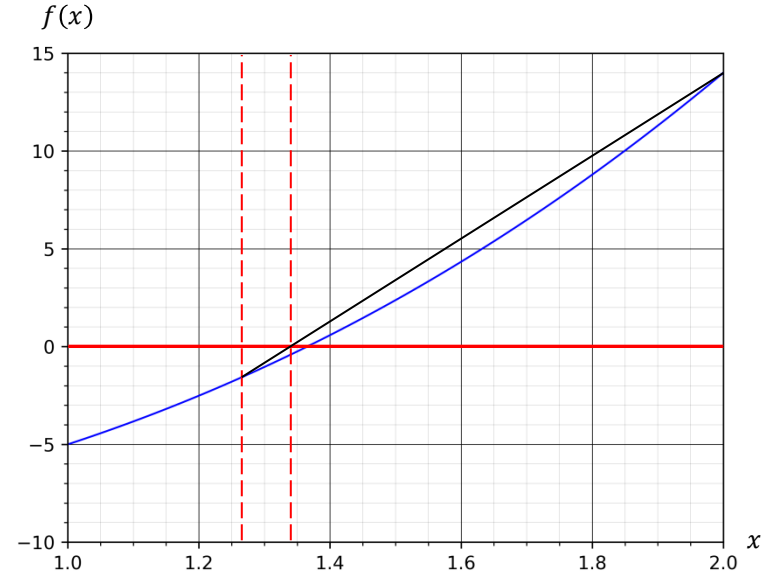
$k = 1$

$$x^{(-1)} = 1, x^{(0)} = 2 \rightarrow x^{(1)} = 2 - \frac{2 - 1}{14 - (-5)} \cdot (14) = 1.263158$$

$$\varepsilon^{(1)} = \frac{|x^{(1)} - x^{(0)}|}{|x^{(1)}|} = \frac{|1.263158 - 2|}{|1.263158|} = 0.583333 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$x^{(0)} = 2, x^{(1)} = 1.263158 \rightarrow x^{(2)} = 1.338828$$



Secant Method

Example 02-05: Find a root of equation $x^3 + 4x^2 - 10 = 0$ using secant method with initial guess $x^{(-1)} = 1, x^{(0)} = 2$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$x^{(k)} = x^{(k-1)} - \frac{x^{(k-1)} - x^{(k-2)}}{f(x^{(k-1)}) - f(x^{(k-2)})} \cdot f(x^{(k-1)}); \quad f(x) = x^3 + 4x^2 - 10$$

$k = 1$

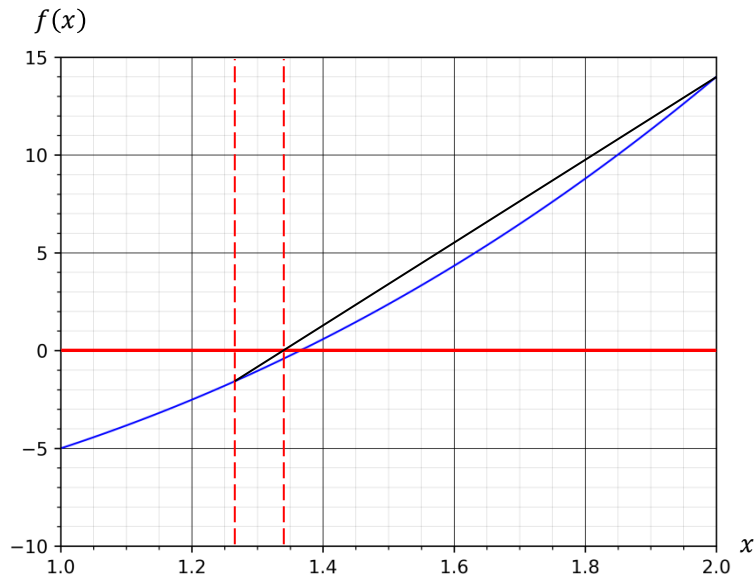
$$x^{(-1)} = 1, x^{(0)} = 2 \rightarrow x^{(1)} = 2 - \frac{2 - 1}{14 - (-5)} \cdot (14) = 1.263158$$

$$\varepsilon^{(1)} = \frac{|x^{(1)} - x^{(0)}|}{|x^{(1)}|} = \frac{|1.263158 - 2|}{|1.263158|} = 0.583333 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$x^{(0)} = 2, x^{(1)} = 1.263158 \rightarrow x^{(2)} = 1.338828$$

$$\varepsilon^{(2)} = \frac{|x^{(2)} - x^{(1)}|}{|x^{(2)}|} = \frac{|1.338828 - 1.263158|}{|1.338828|} = 0.056520 > \varepsilon_t$$



Secant Method

Example 02-05: Find a root of equation $x^3 + 4x^2 - 10 = 0$ using secant method with initial guess $x^{(-1)} = 1, x^{(0)} = 2$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$x^{(k)} = x^{(k-1)} - \frac{x^{(k-1)} - x^{(k-2)}}{f(x^{(k-1)}) - f(x^{(k-2)})} \cdot f(x^{(k-1)}); \quad f(x) = x^3 + 4x^2 - 10$$

$k = 1$

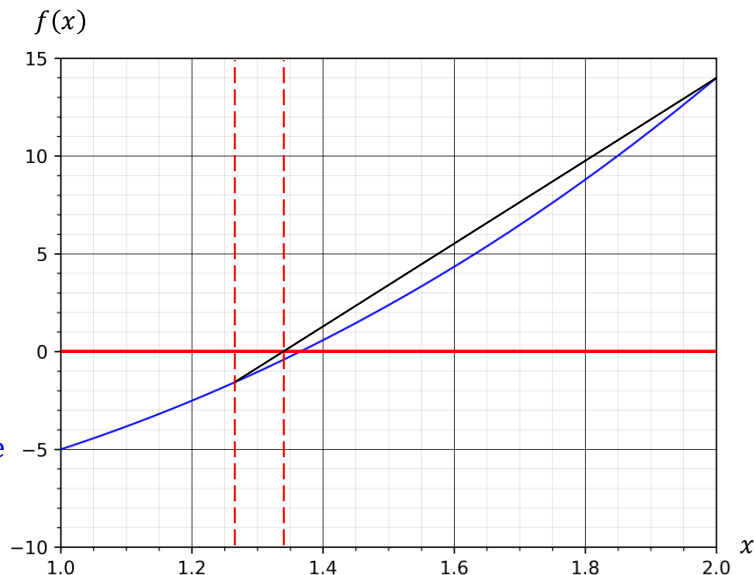
$$x^{(-1)} = 1, x^{(0)} = 2 \rightarrow x^{(1)} = 2 - \frac{2 - 1}{14 - (-5)} \cdot (14) = 1.263158$$

$$\varepsilon^{(1)} = \frac{|x^{(1)} - x^{(0)}|}{|x^{(1)}|} = \frac{|1.263158 - 2|}{|1.263158|} = 0.583333 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$x^{(0)} = 2, x^{(1)} = 1.263158 \rightarrow x^{(2)} = 1.338828$$

$$\varepsilon^{(2)} = \frac{|x^{(2)} - x^{(1)}|}{|x^{(2)}|} = \frac{|1.338828 - 1.263158|}{|1.338828|} = 0.056520 > \varepsilon_t \rightarrow \text{continue}$$



Secant Method

Example 02-05: Find a root of equation $x^3 + 4x^2 - 10 = 0$ using secant method with initial guess $x^{(-1)} = 1, x^{(0)} = 2$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$x^{(k)} = x^{(k-1)} - \frac{x^{(k-1)} - x^{(k-2)}}{f(x^{(k-1)}) - f(x^{(k-2)})} \cdot f(x^{(k-1)}); \quad f(x) = x^3 + 4x^2 - 10$$

$k = 1$

$$x^{(-1)} = 1, x^{(0)} = 2 \rightarrow x^{(1)} = 2 - \frac{2 - 1}{14 - (-5)} \cdot (14) = 1.263158$$

$$\varepsilon^{(1)} = \frac{|x^{(1)} - x^{(0)}|}{|x^{(1)}|} = \frac{|1.263158 - 2|}{|1.263158|} = 0.583333 > \varepsilon_t \rightarrow \text{continue}$$

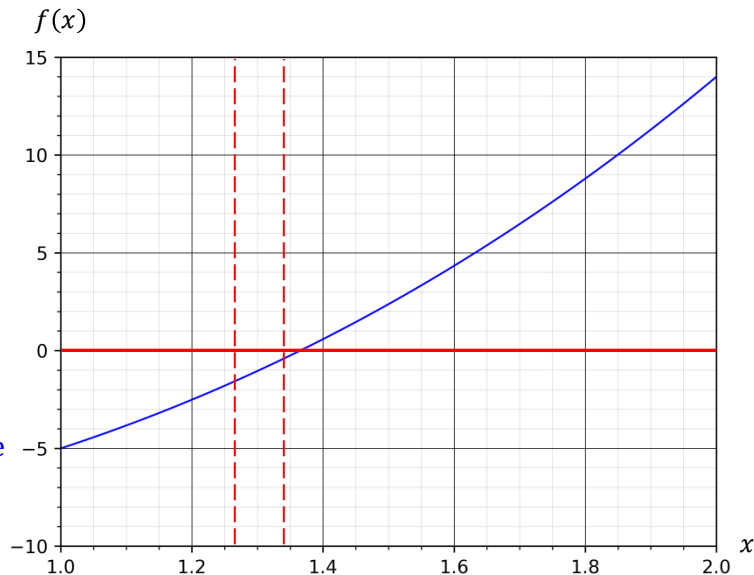
$k = 2$

$$x^{(0)} = 2, x^{(1)} = 1.263158 \rightarrow x^{(2)} = 1.338828$$

$$\varepsilon^{(2)} = \frac{|x^{(2)} - x^{(1)}|}{|x^{(2)}|} = \frac{|1.338828 - 1.263158|}{|1.338828|} = 0.056520 > \varepsilon_t \rightarrow \text{continue}$$

$k = 3$

$$x^{(1)} = 1.263158, x^{(2)} = 1.338828$$



Secant Method

Example 02-05: Find a root of equation $x^3 + 4x^2 - 10 = 0$ using secant method with initial guess $x^{(-1)} = 1, x^{(0)} = 2$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$x^{(k)} = x^{(k-1)} - \frac{x^{(k-1)} - x^{(k-2)}}{f(x^{(k-1)}) - f(x^{(k-2)})} \cdot f(x^{(k-1)}); \quad f(x) = x^3 + 4x^2 - 10$$

$k = 1$

$$x^{(-1)} = 1, x^{(0)} = 2 \rightarrow x^{(1)} = 2 - \frac{2 - 1}{14 - (-5)} \cdot (14) = 1.263158$$

$$\varepsilon^{(1)} = \frac{|x^{(1)} - x^{(0)}|}{|x^{(1)}|} = \frac{|1.263158 - 2|}{|1.263158|} = 0.583333 > \varepsilon_t \rightarrow \text{continue}$$

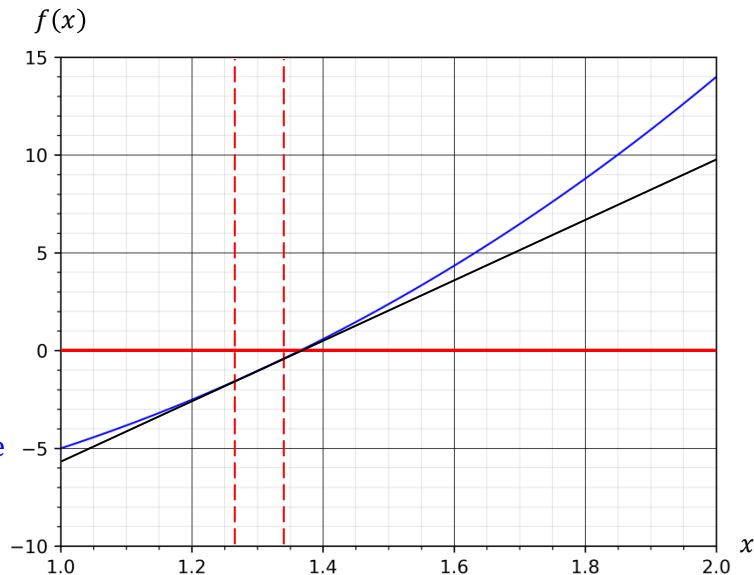
$k = 2$

$$x^{(0)} = 2, x^{(1)} = 1.263158 \rightarrow x^{(2)} = 1.338828$$

$$\varepsilon^{(2)} = \frac{|x^{(2)} - x^{(1)}|}{|x^{(2)}|} = \frac{|1.338828 - 1.263158|}{|1.338828|} = 0.056520 > \varepsilon_t \rightarrow \text{continue}$$

$k = 3$

$$x^{(1)} = 1.263158, x^{(2)} = 1.338828$$



Secant Method

Example 02-05: Find a root of equation $x^3 + 4x^2 - 10 = 0$ using secant method with initial guess $x^{(-1)} = 1, x^{(0)} = 2$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$x^{(k)} = x^{(k-1)} - \frac{x^{(k-1)} - x^{(k-2)}}{f(x^{(k-1)}) - f(x^{(k-2)})} \cdot f(x^{(k-1)}); \quad f(x) = x^3 + 4x^2 - 10$$

$k = 1$

$$x^{(-1)} = 1, x^{(0)} = 2 \rightarrow x^{(1)} = 2 - \frac{2 - 1}{14 - (-5)} \cdot (14) = 1.263158$$

$$\varepsilon^{(1)} = \frac{|x^{(1)} - x^{(0)}|}{|x^{(1)}|} = \frac{|1.263158 - 2|}{|1.263158|} = 0.583333 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$x^{(0)} = 2, x^{(1)} = 1.263158 \rightarrow x^{(2)} = 1.338828$$

$$\varepsilon^{(2)} = \frac{|x^{(2)} - x^{(1)}|}{|x^{(2)}|} = \frac{|1.338828 - 1.263158|}{|1.338828|} = 0.056520 > \varepsilon_t \rightarrow \text{continue}$$

$k = 3$

$$x^{(1)} = 1.263158, x^{(2)} = 1.338828$$

